



TELCO ANALYTICS



ERD



SQL



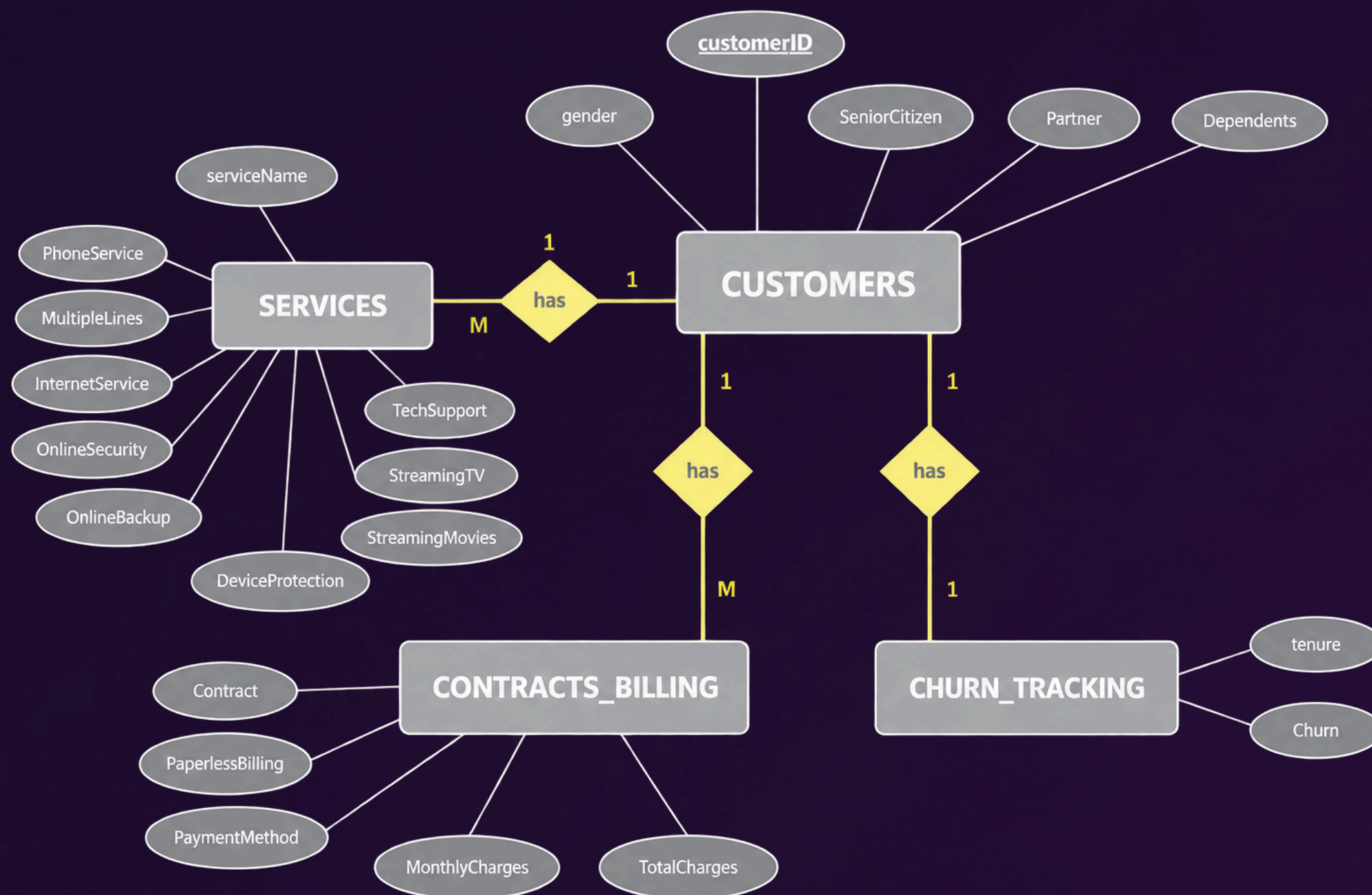
Payton



AI Model



Dashboard



Database Design & Architecture Summary

- **Schema Setup:** Created TelecomDB in SQL Server, divided into four core tables: CUSTOMERS, SERVICES, CONTRACTS_BILLING, and CHURN_TRACKING.
- **Core Relationships:** CUSTOMERS serves as the primary entity (customerID PK), linked via Foreign Keys to SERVICES (**1:M**), CONTRACTS_BILLING (**1:M**), and CHURN_TRACKING (**1:1**).
- **Data Pipeline:** Executed schema creation via DDL and populated tables directly from clean_telecom_data using INSERT INTO ... SELECT.



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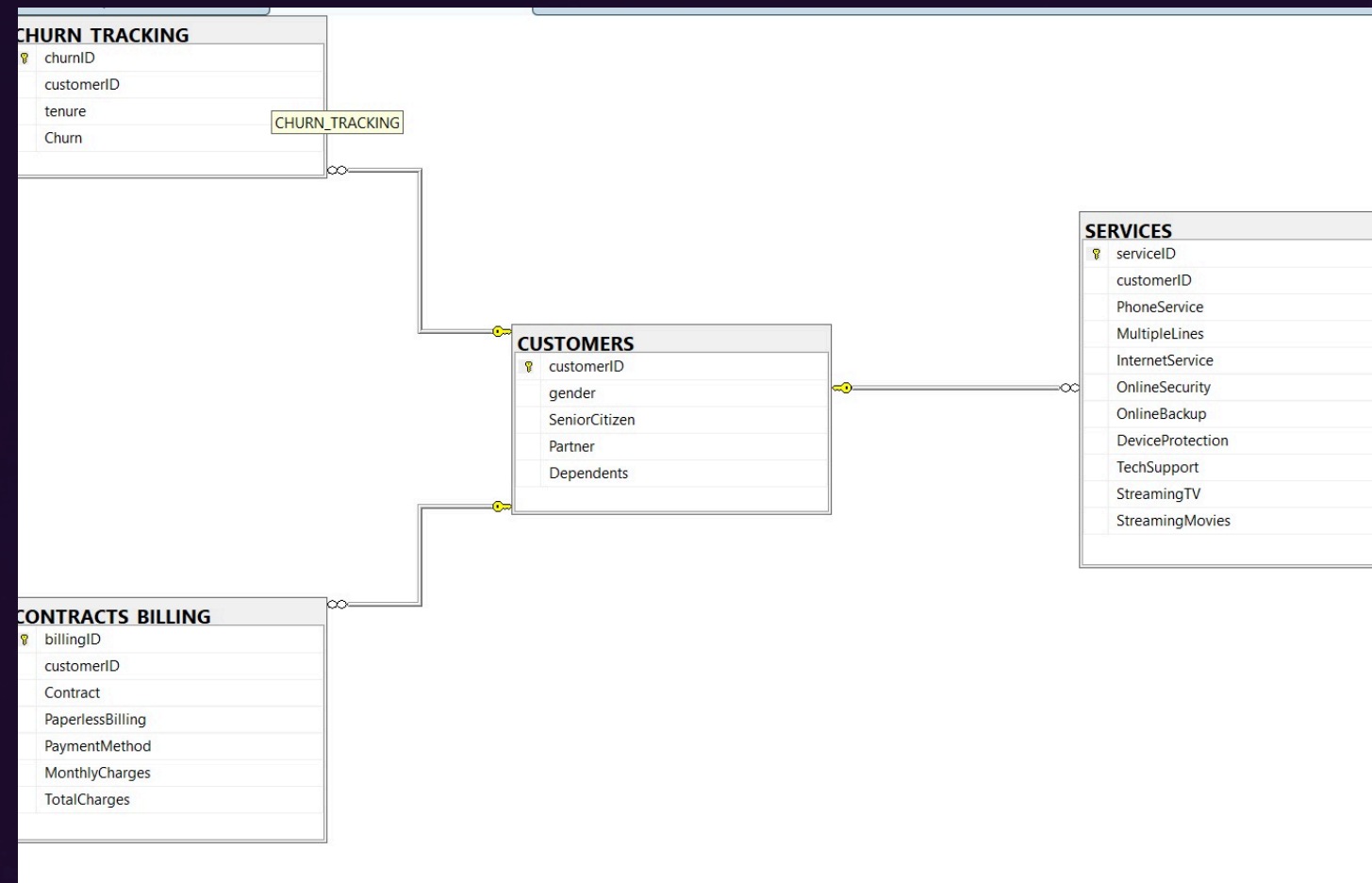
Python



AI Model



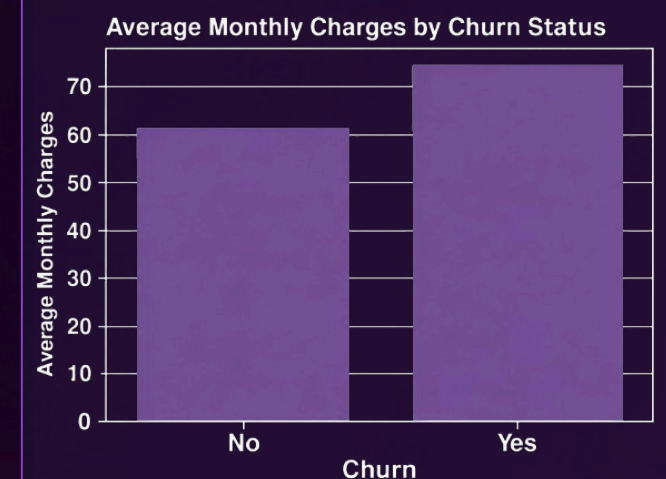
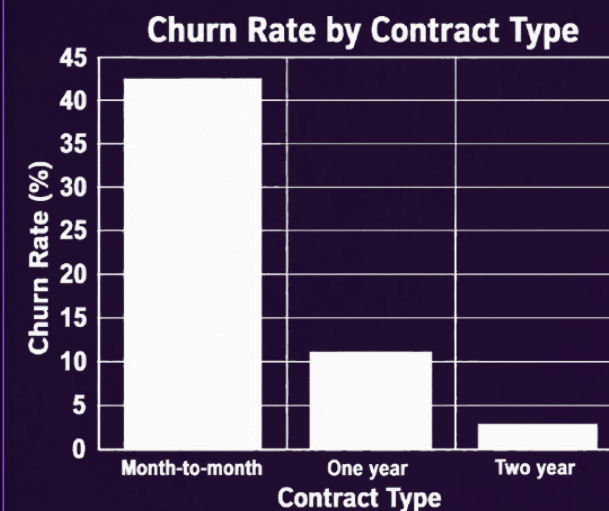
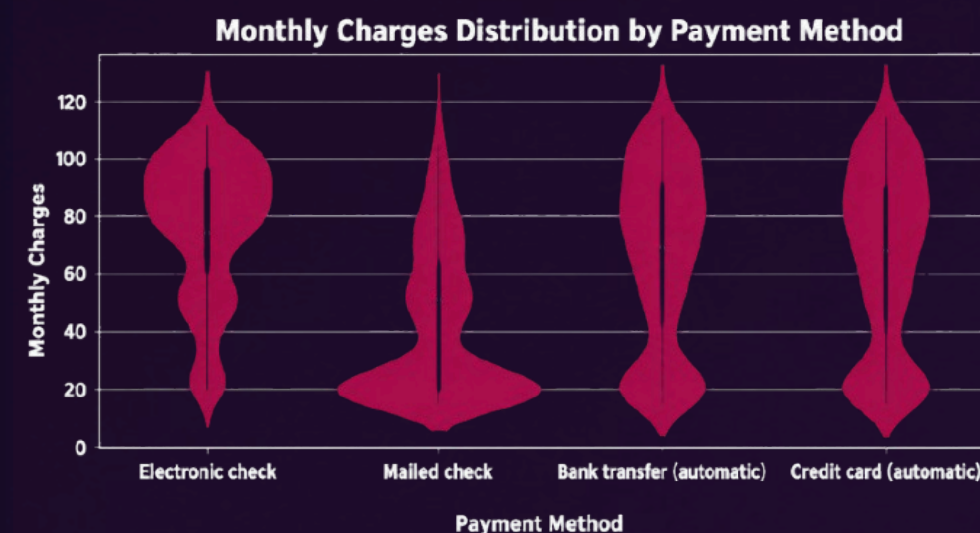
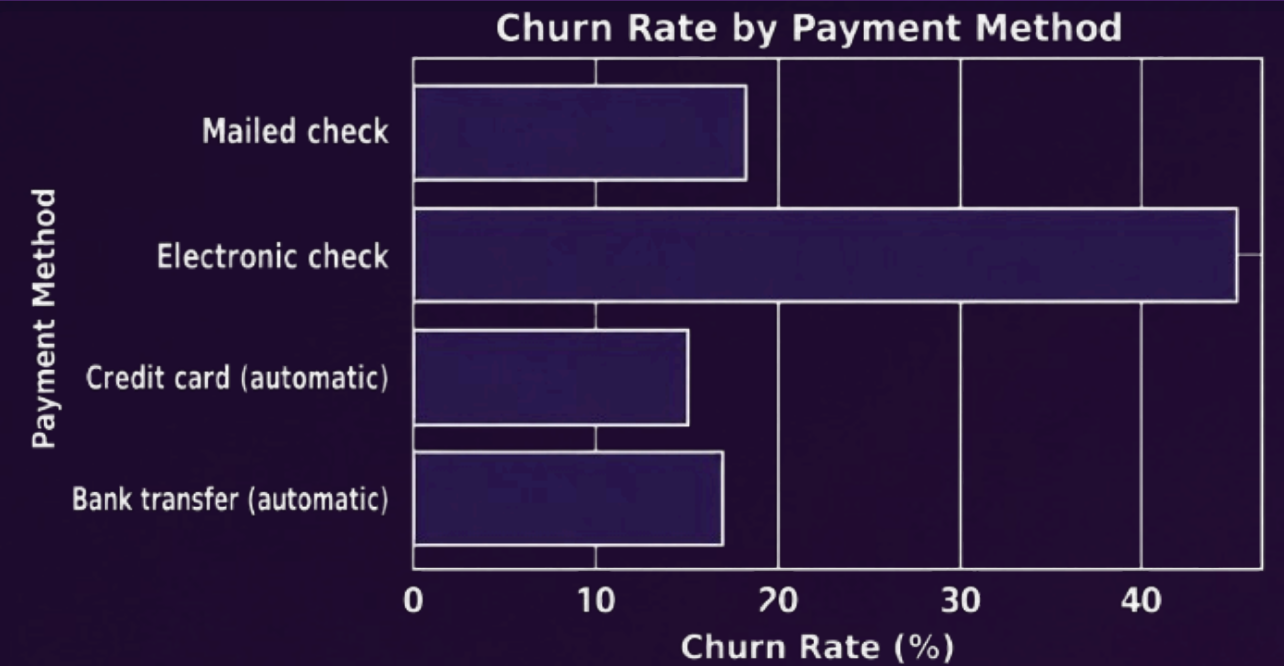
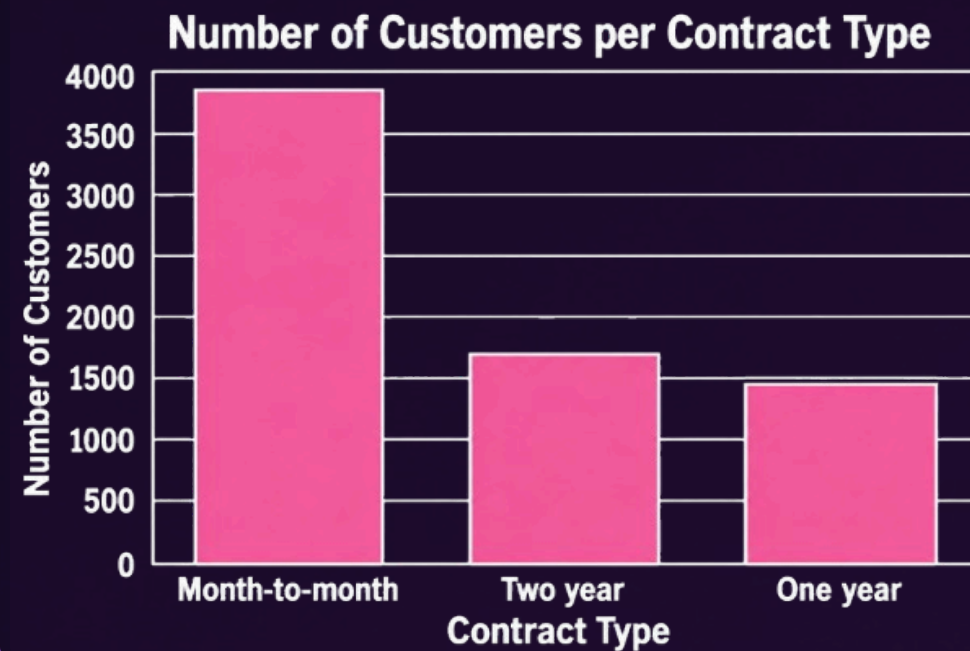
Dashboard



Technical Overview & Business Impact

- **Data Cleaning & Preprocessing:** Handled missing values, standardized text formats, and eliminated duplicate records using Common Table Expressions (CTEs) and ROW_NUMBER().
- **KPI & Feature Analytics:** Extracted key retention metrics, average billing patterns, and contract distribution across segments using SQL Window Functions.
- **Database Normalization (ERD):** Restructured the flat dataset into a normalized 4-table relational schema to optimize query performance and eliminate data redundancy.
- **Business Impact:** Bridged technical data engineering with business analytics to provide actionable insights that empower management to mitigate customer churn.

Access the complete code and
documentation here:  **GITHUB**



Key Analytical Focus Areas

- **Contract Types:** Impact of short-term vs. long-term commitments on churn.
- **Support Services:** Effect of technical support availability on retention.
- **Billing & Payments:** Relationship between monthly charges, payment options, and customer loss.

Business Value

- Early identification of at-risk customers.
- Data-driven retention and loyalty strategies.

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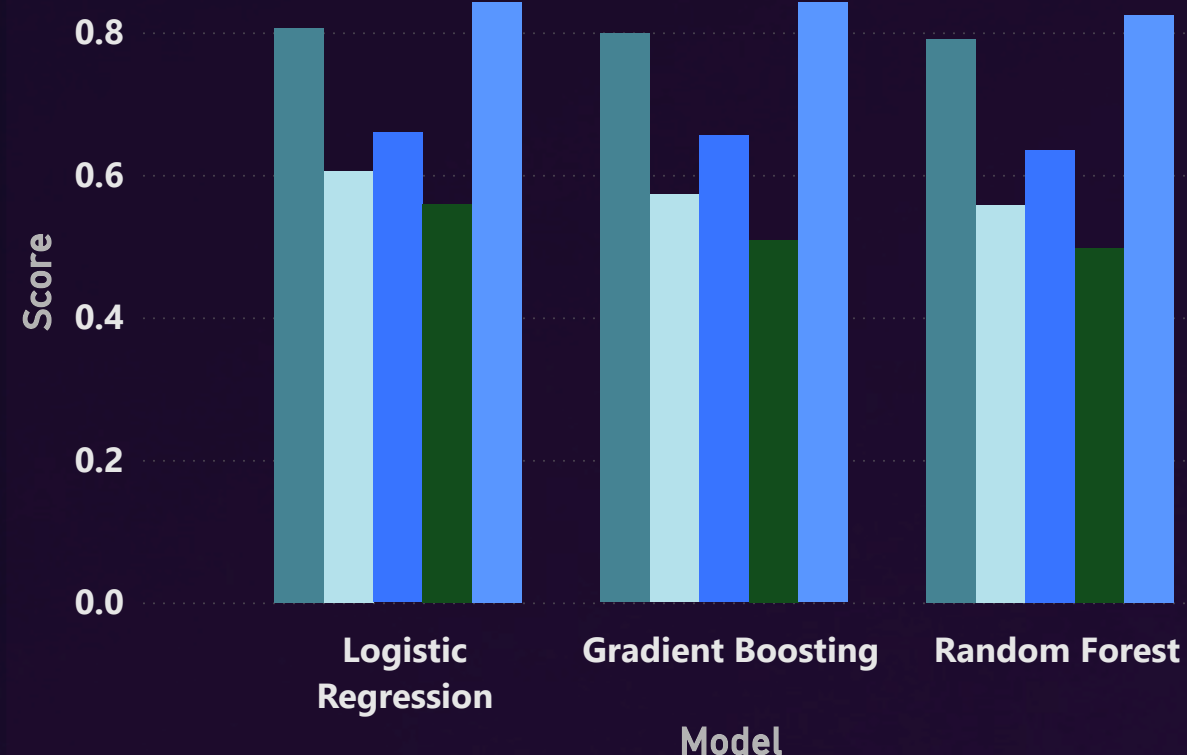
AI Model



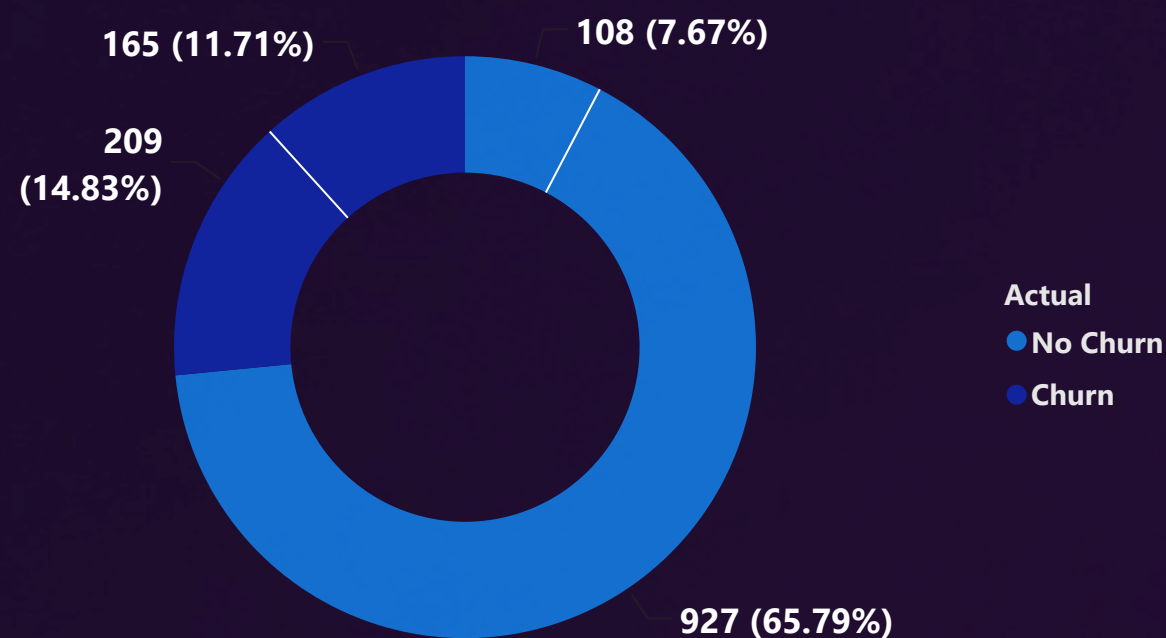
Dashboard



● Sum of Accuracy ● Sum of F1-Score ● Sum of Precision ● Sum of Recall ● Sum of ROC-AUC



Sum of Value by Actual and Predicted



Model Performance Summary

- **Top Model:** Logistic Regression achieves the highest performance with **80.6% Accuracy**.
- **Key Metrics:** Leads across Precision and F1-Score compared to Gradient Boosting and Random Forest.
- **Recommendation:** Best suited model for deployment to predict customer churn reliably.

Logistic Regression Matrix Insights

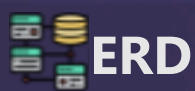
- **High Accuracy:** Correctly identified **927 Retained** and **209 Churned** customers.
- **Low Error Rate:** Only **165 false negatives** (churn risks missed by the model).

Access the complete code and documentation here:





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7.043K

Total customers



\$64.76

MonthlyCharges



32.37

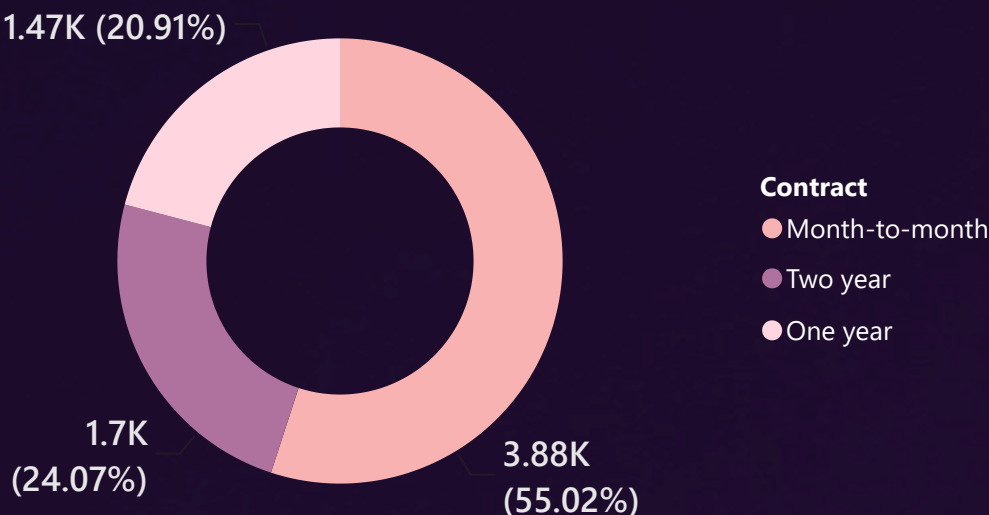
Avg of tenure



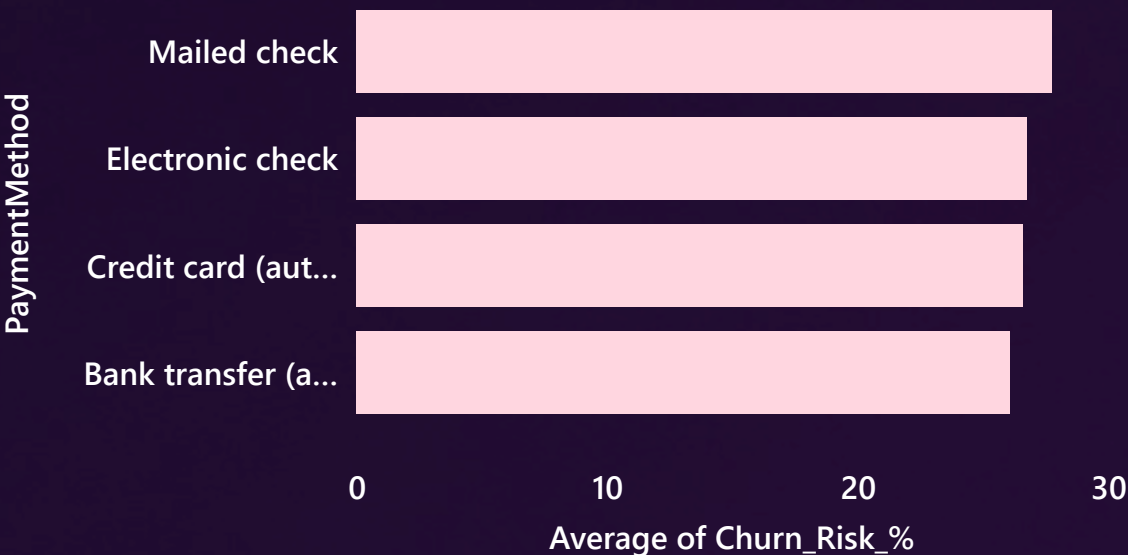
26.78

Churn_Risk_%

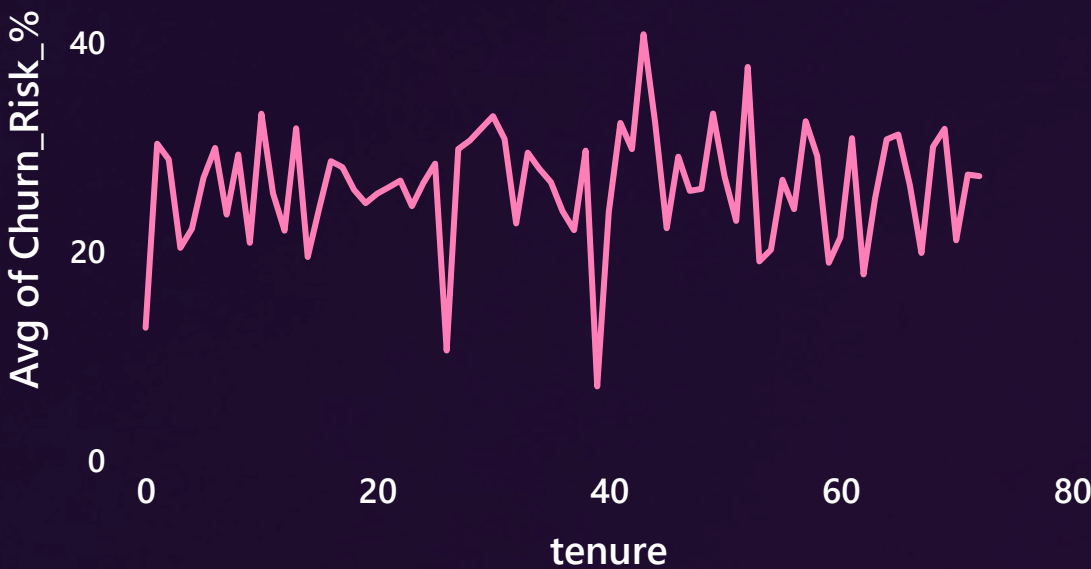
Customers By Contract Type



Churn_Risk by PaymentMethod



Avg of Churn_Risk_% by tenure



Count of customerID by OnlineSecurity and Churn

